Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · AI Agent Development · Eval Frameworks · Guardrails · Prompt Management · Model Routing · Fintech · Python · AWS

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SUMMARY

Senior software engineer specializing in **AI agent development, eval frameworks, guardrail systems, and observability** for production LLM-powered products. Currently at Intuit building Anthropic Claude integrations into production financial services: **agent architectures** with tool calling, prompt strategies, and structured outputs; **eval frameworks** with LLM-as-judge scoring, golden datasets, and CI regression gating; **guardrails** with input validation, sensitive-data handling, and fallbacks; production monitoring via structured logging and CloudWatch quality signals. Built **AgentProbe** - an open-source LLM eval and observability platform with **model routing** across 6 providers (Anthropic, OpenAI, Azure OpenAI, AWS Bedrock, Groq, Ollama), cost-aware routing logic, CI regression harness, and full per-run tracing. Built **Leasel** - a production agentic platform (LangGraph, 8-agent orchestration) with durable context management, prompt-driven conditional routing, HITL guardrails, and a CI eval harness (93.6% precision, 1,240 test cases). At Entrata (7+ years, 100K+ users), owned mission-critical async distributed systems and compliance data pipelines in regulated multi-tenant SaaS. Python, FastAPI, PostgreSQL, AWS, Docker, Kubernetes. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

AI Agent Development · Agent Architectures · Orchestration Patterns
Prompt Strategies · Prompt Management · Structured Outputs · Tool Calling
Eval Frameworks · LLM-as-Judge · Golden Datasets · CI Regression
Guardrail Systems · Safety Controls · HITL · Fallbacks
Observability · Tracing · Structured Logging · Production Signals
Model Routing · Multi-Provider LLM · Cost-Aware Routing
Fintech · Financial Services AI · Member-Facing Products
Python · FastAPI · PostgreSQL · Redis · Backend Engineering
AWS (Lambda · Bedrock · SQS · S3) · Docker · Kubernetes
React · TypeScript · Full-Stack Development
High-Scale Distributed Systems · Async Processing · Idempotency
TDD · Code Reviews · Mentorship · Agile Engineering

EXPERIENCE

Software Engineer

Jun 2026 - Present

Intuit — Virginia, USA (Remote) — Financial Services AI

- **AI agent development in production fintech:** Build LangChain-based agent architectures integrating Anthropic Claude into production financial services - prompt strategies with tool calling and retrieval-grounded context, structured-output schemas, orchestration patterns for multi-step financial workflows; guardrail systems with input validation, sensitive-data handling, and fallbacks for safe member-facing behavior.
- **Eval frameworks and observability:** Build LLM-as-judge eval frameworks with golden datasets and CI regression gates; monitor production AI via structured logging and CloudWatch quality signals; perform qualitative error analysis on failures and convert them into eval coverage. Build prompt management patterns and playbooks for team reuse.
- Build ML data pipelines (scikit-learn, XGBoost); React/TypeScript front ends; deploy via Docker + GitHub Actions CI/CD to AWS with full-stack ownership.

Senior Software Engineer

Oct 2020 - Jul 2024

Entrata — Apartment & Affordable Housing SaaS (regulated multi-tenant, 100K+ monthly active users)

- Owned mission-critical data pipelines and async distributed systems end-to-end - HUD compliance ETL (180K records, zero violations across 6 years), RabbitMQ event-driven processing with idempotency guarantees and dead-letter routing, PHI-compliant AWS S3 migration (100% coverage, zero incidents).
- Led Docker microservices redesign (50% scalability improvement); optimized 200+ SQL queries on PostgreSQL schemas (30% performance gain); Python/Bash automation eliminating 80% of manual workload; mentored 5+ engineers; 300+ code reviews across 120+ Agile sprints.

Junior Software Engineer

Sep 2018 - Oct 2020

Entrata India — Pune

- Built backend modules (Python, PHP, Java, PostgreSQL); delivered large-scale data integration workflows (1.6M records); led AWS S3 migration from legacy storage.

PROJECTS

AgentProbe - LLM Eval Framework, Observability & Model Routing Platform

2025 | Python, FastAPI, PostgreSQL, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- Open-source **eval framework and observability platform**: offline golden-dataset evals, LLM-as-judge scoring, CI regression harness gating releases on eval thresholds, failure handling with structured error classification; **cost-aware model routing** across Anthropic, OpenAI, Azure OpenAI, AWS Bedrock, Groq, and Ollama; full per-run tracing and structured logging. Used as a production release gate before every AI deploy.

Leasel - AI Agent Platform: Orchestration · Prompt Strategies · HITL Guardrails

2026 | Python, LangGraph (LangChain), FastAPI, ChromaDB, MCP, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production AI agent platform: **8-agent LangGraph orchestration** with tool calling, durable context management, prompt-driven conditional routing; **guardrail system** routing regulated decisions to humans for safety; **hybrid search RAG** (OCR ingestion, TF-IDF + semantic reranking, citation grounding for hallucination mitigation); **CI eval harness** (LLM-as-judge, golden datasets: 93.6% precision, 87.9% F1 across 1,240 test cases).

AI on FHIR — Conversational AI & API Integration

2024 | Python, Flask, React, BERT, LLMs, FHIR REST APIs, Docker | github.com/venkatalakshmiparimala/AI_on_FHIR

- Translates natural-language questions into FHIR REST API calls using LLMs, BERT, and embeddings - cut clinical lookup time ~60%. OAuth-secured API integration, HIPAA-style RBAC, Docker-containerized. Full-stack: Python/Flask backend + React frontend.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia I GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017